Local Differential Privacy: Refined Mechanism Design and Utility Analysis

Ye Zheng

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Committee: Dr. Sumita Mishra, Dr. Haibo Yang, Dr. Weijie Zhao

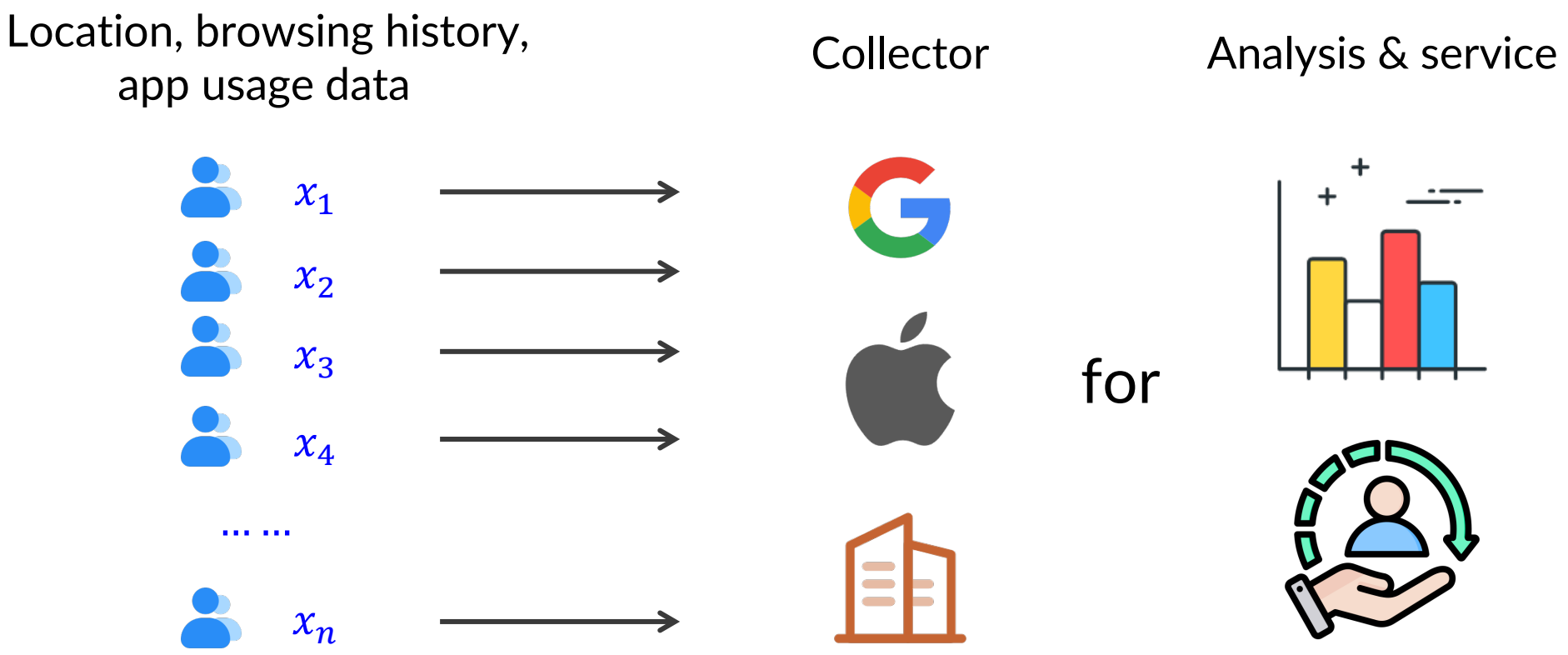
RIT | Rochester Institute of Technology

PDF, slides, ... 👉 https://zhengyeah.com

Data Collection Everywhere

RIT

- Users' personal data are collected by companies for analysis or user service



Users' Data Privacy

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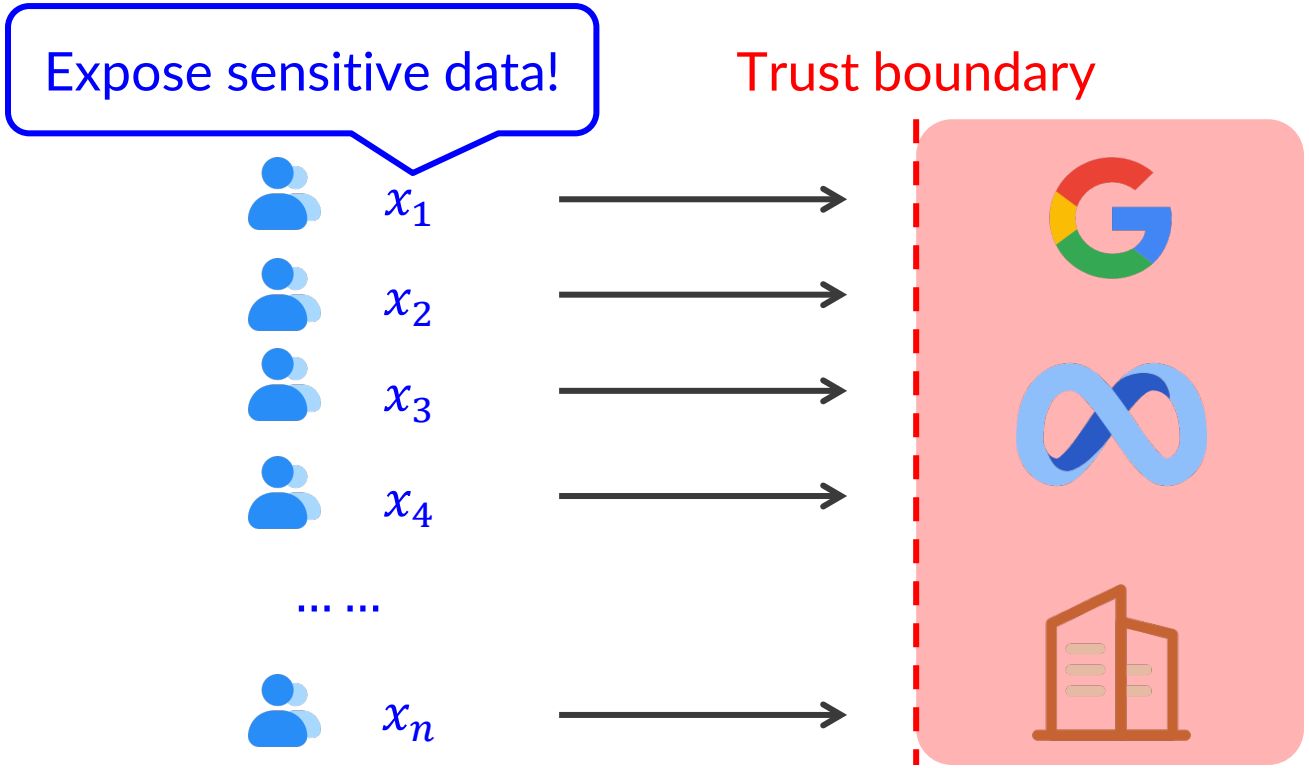
  - these companies may be untrusted to collect users' sensitive data



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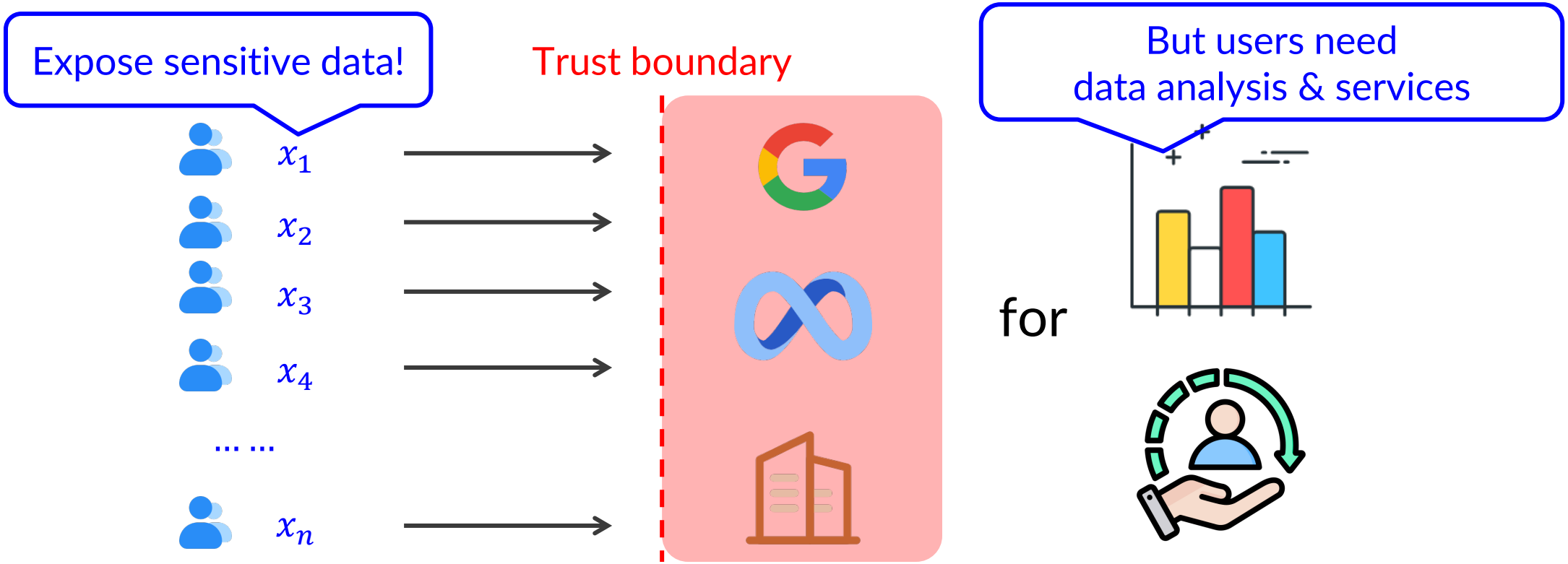

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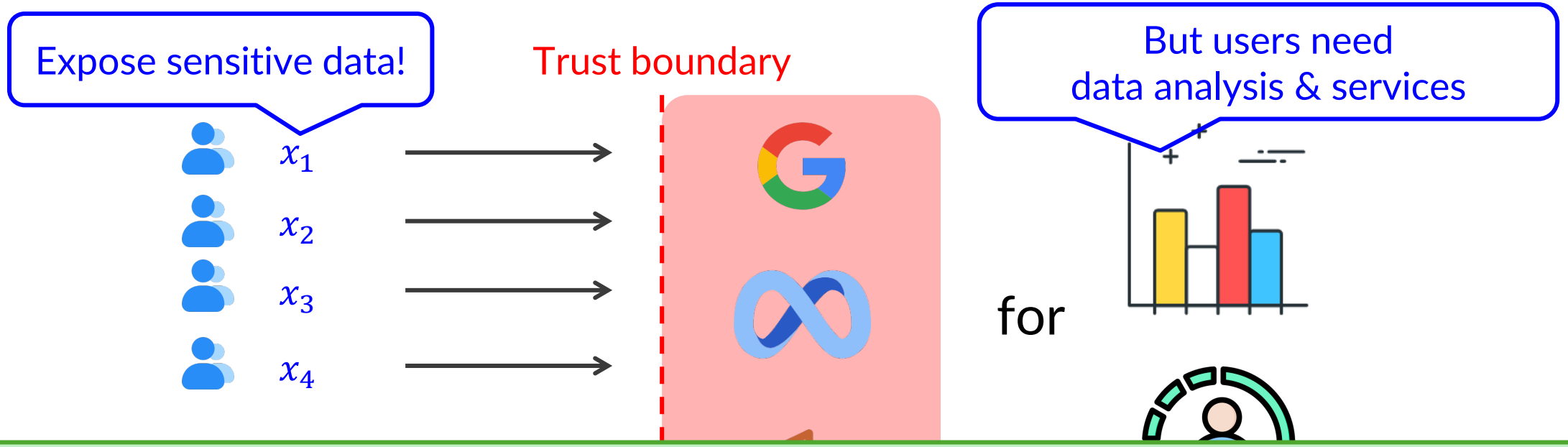
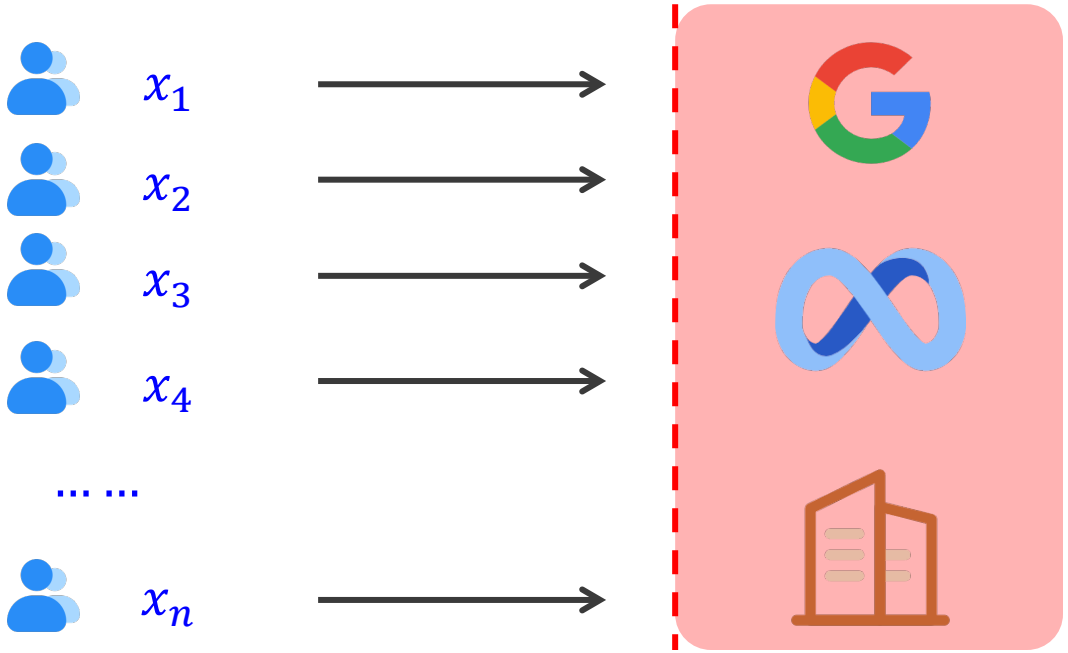


Q: How to provide data analysis & services for users **while** protecting their data privacy?

Privacy-Preserving Computation

RIT

- Users' personal data are collected by companies for analysis or user service
  - these companies may be untrusted to collect users' sensitive data
  - how to compute $f(x_1, x_2, \dots, x_n)$ without revealing $x_1, x_2, \dots, x_n$?



for $f(x_1, x_2, \dots, x_n)$

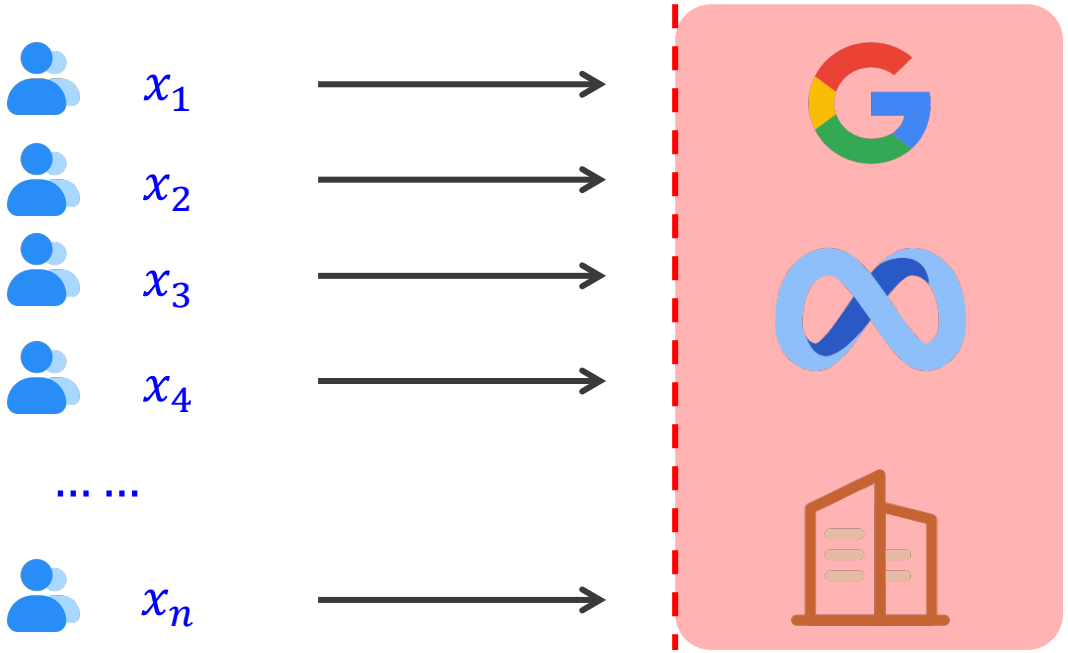
Privacy-Preserving Computation - **Techniques**

- Homomorphic encryption (HE), multi-party computation (MPC), local differential privacy (LDP), etc

Privacy-Preserving Computation - **HE**

RIT

- Homomorphic encryption (HE):
  - "homomorphic": keeping structure
  - design algorithm $\{Enc, Dec\} \rightarrow Dec\big(f(Enc(x_1), Enc(x_2), \dots, Enc(x_n))\big) = f(x_1, x_2, \dots, x_n)$



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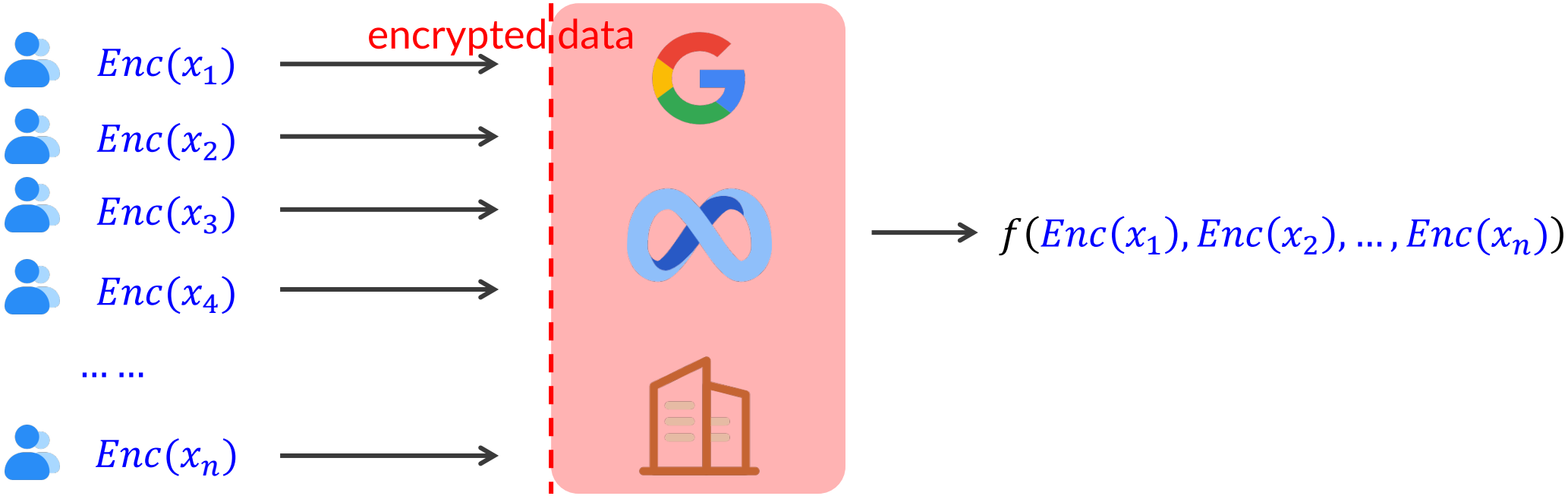

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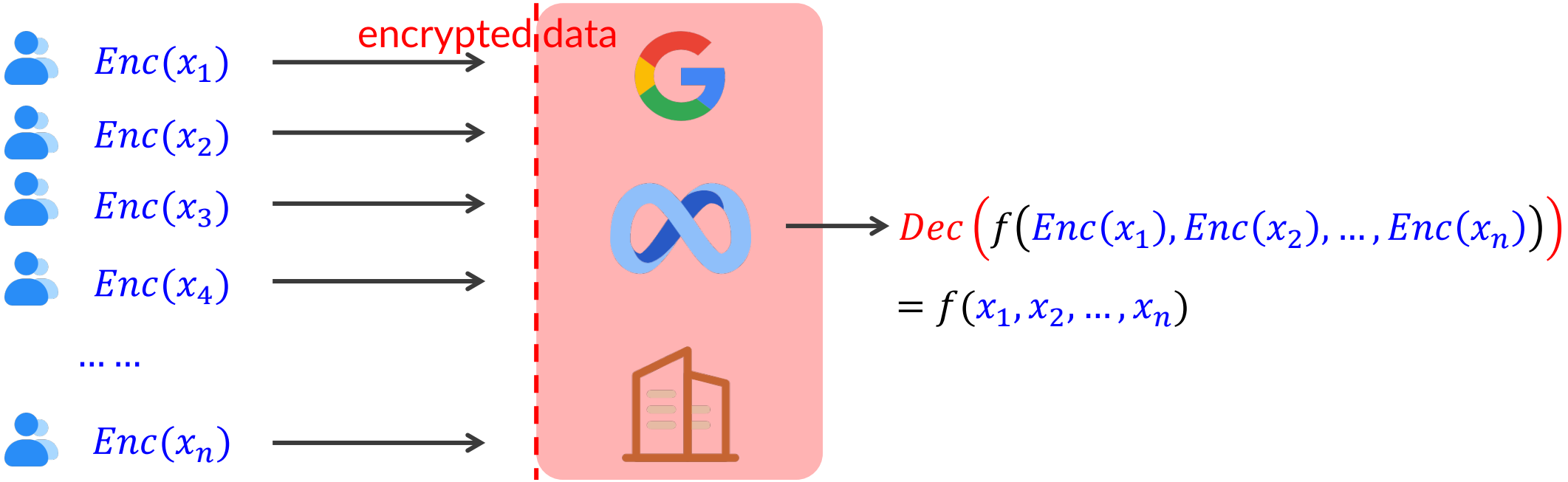

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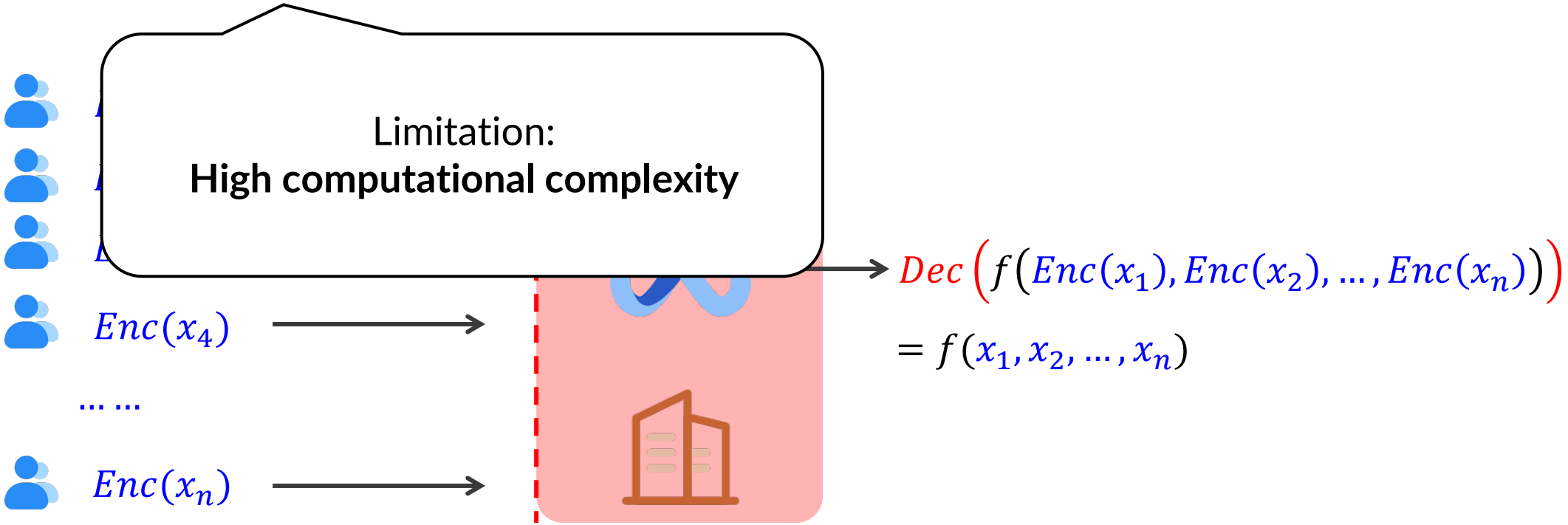

Privacy-Preserving Computation - **MPC**

RIT

- Multi-party computation (MPC):
  - no central party
  - jointly compute f without revealing x_i
- Example: $f(x_1, x_2) = x_1 + x_2$

x_1

for $f(x_1, x_2)$

x_2

… …

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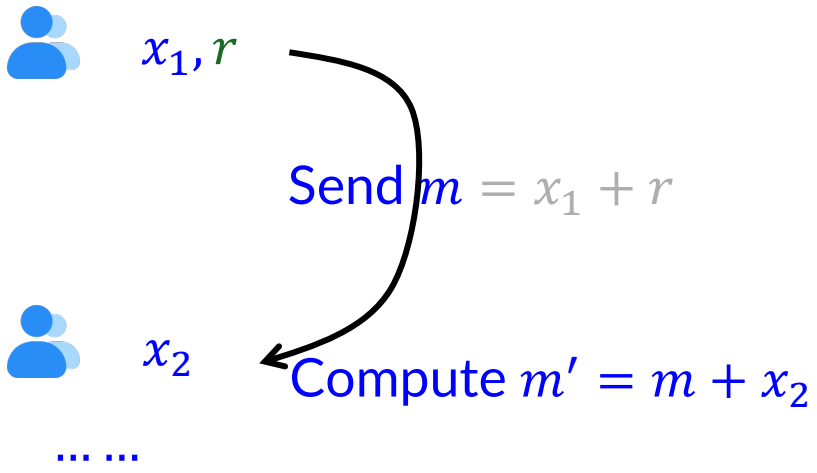

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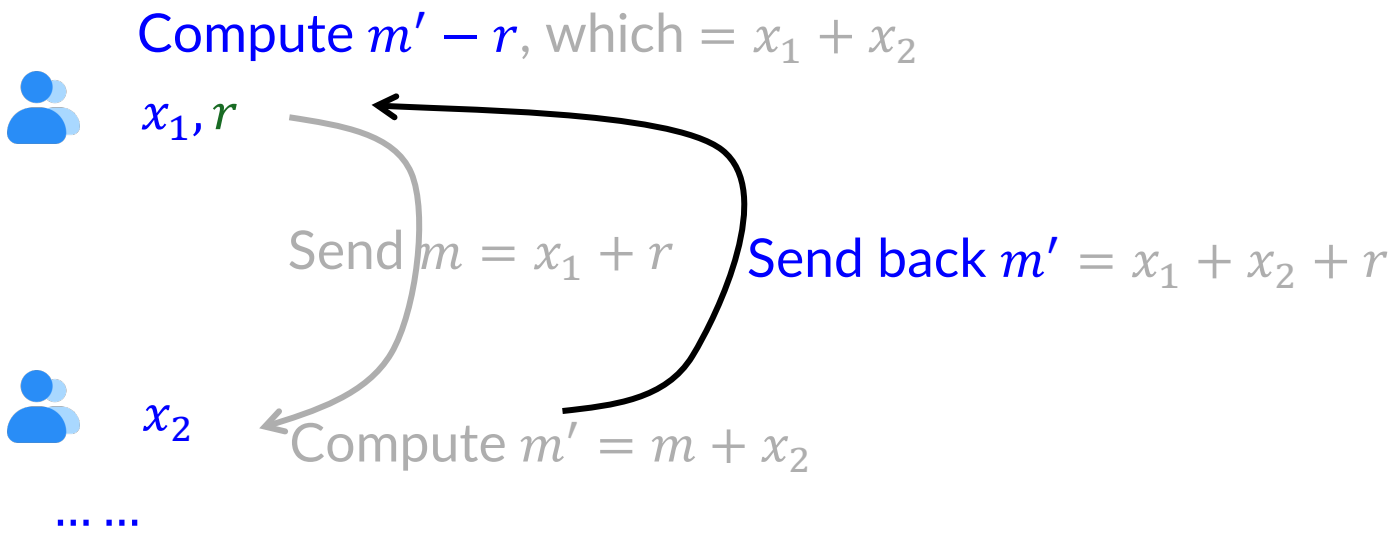

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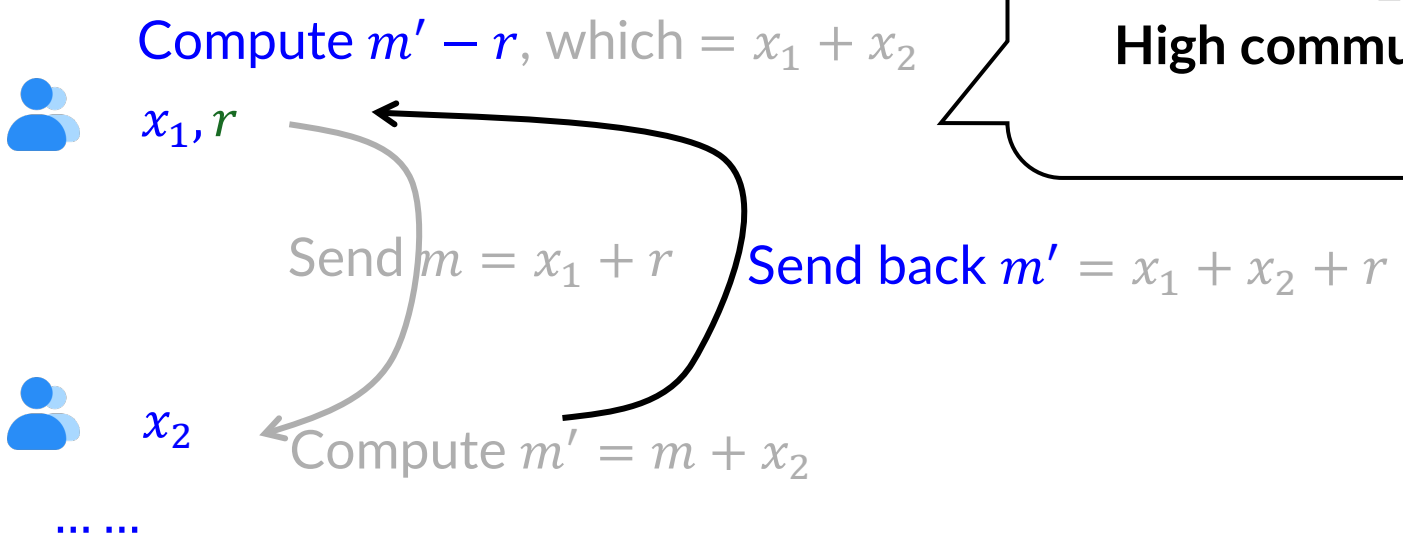

Limitation:
High communication complexity

Privacy-Preserving Computation - **LDP**

RIT

- Local differential privacy (LDP):
  - hard to differentiate the sensitive data from other data
  - each user locally perturbs x_i to $\tilde{x}_i$ $\rightarrow$ $f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n) \approx f(x_1, x_2, \dots, x_n)$

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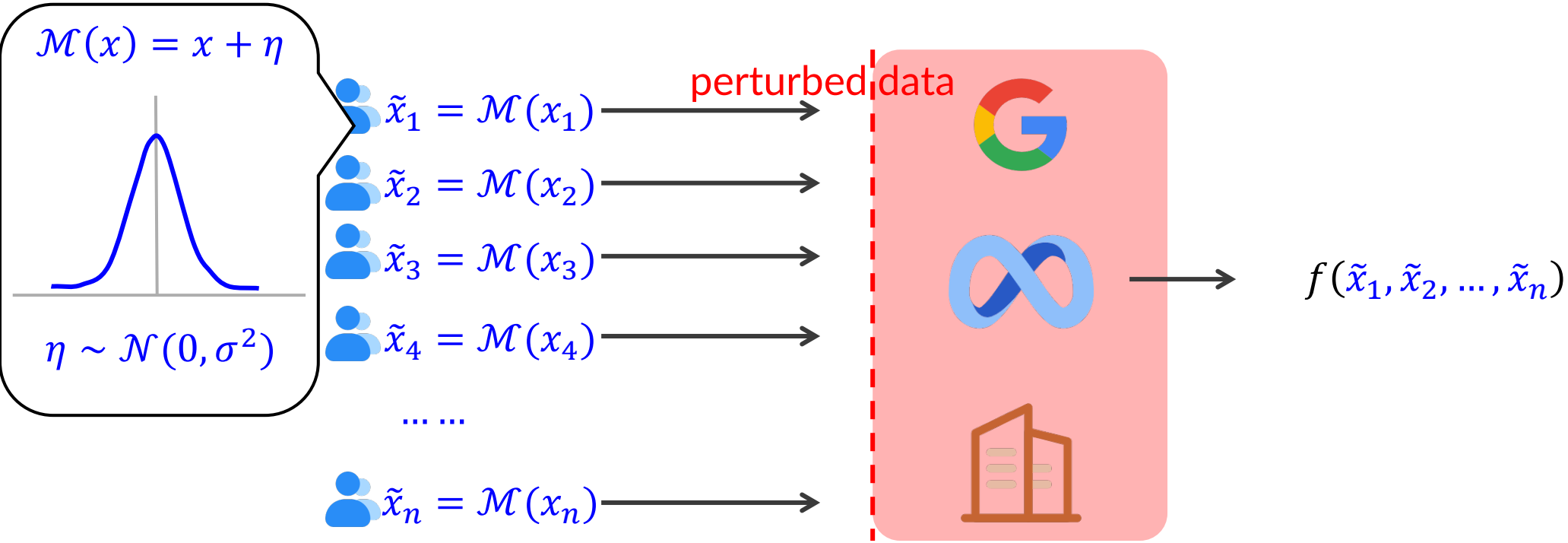

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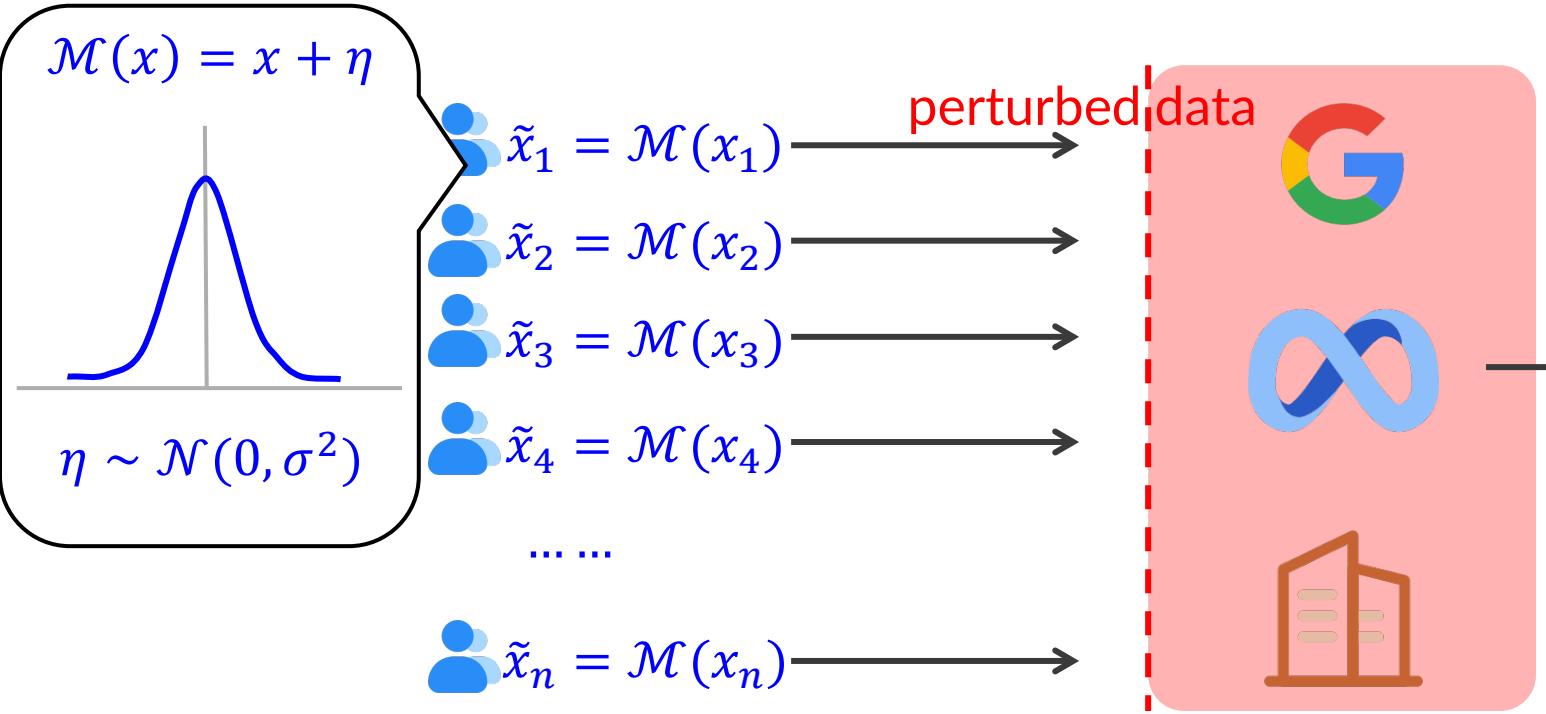


$$f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$$

$$\mathrm{E}[f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)] = f(x_1, x_2, \dots, x_n)$$

Privacy-Preserving Computation - **LDP**

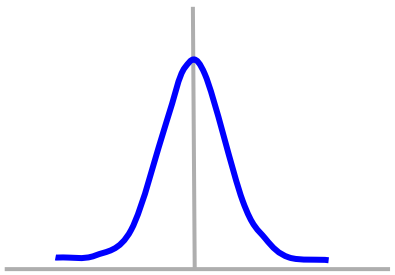
RIT

- Local differential privacy (L
  - hard to differentiate the
  - each user locally perturb

Advantages:
Negligible computational complexity
No communication between users

But approximated f

$\mathcal{M}(x) = x + \eta$



$\eta \sim \mathcal{N}(0, \sigma^2)$

$\tilde{x}_1 = \mathcal{M}(x_1)$

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$\tilde{x}_3 = \mathcal{M}(x_3)$

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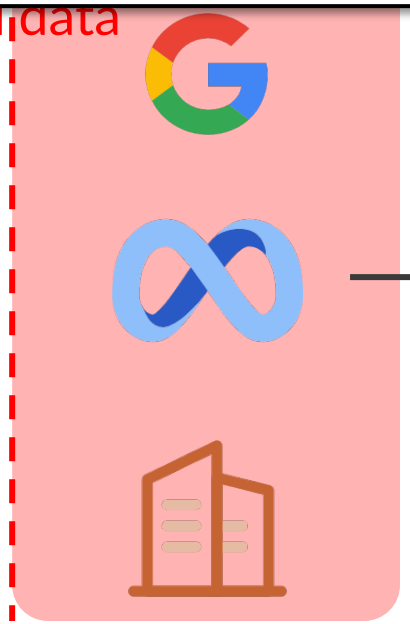
… …

$\tilde{x}_n = \mathcal{M}(x_n)$

perturbed data



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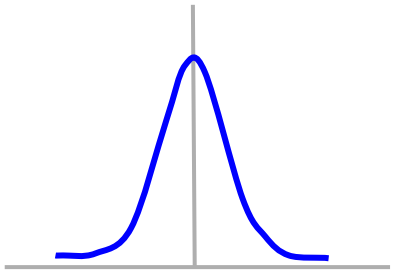
LDP – Applications

RIT

Chrome uses LDP to collect homepage settings, extension usage, etc



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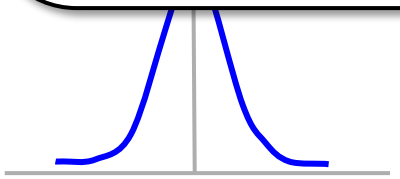
Emoji usage, new word for keyboards, Safari URL statistics, health analytics



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LDP – Formal Privacy

RIT

- After applying $\mathcal{M}$, confidence of distinguishing sensitive x_1 and x_2 from observation $\tilde{x}$:

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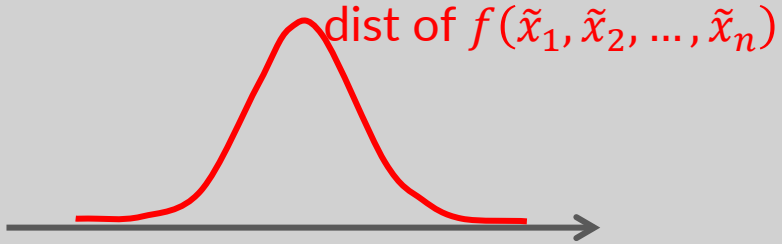
- The collector's / adversary's view: hard to know the sensitive data

Privacy quantified by ε

$x_1 \rightarrow \mathcal{M} \rightarrow \tilde{x}$

Provable defense against data inference attack

Data utility by approximated error



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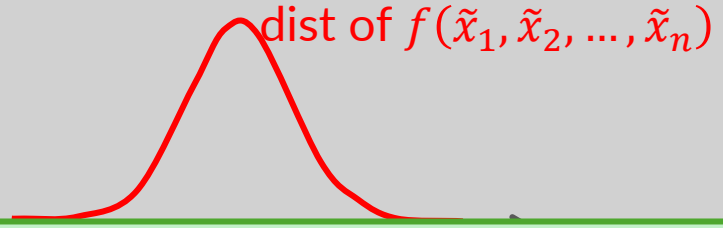

Fundamental direction: **Design** of $\mathcal{M}$ to **optimize** the privacy—utility tradeoff

LDP – Formal Privacy

RIT

Utility analysis of $f \circ \mathcal{M}$

$$f(x_1, x_2, \dots, x_n) \coloneqq \sum_{i=1}^{n} x_i \quad \text{or} \quad f(x_1, x_2, \dots, x_n) \coloneqq \{x_1, x_2, \dots, x_n\} \rightarrow$$ Variance, MSE

$f(x_1, x_2, \dots, x_n) \coloneqq h: \mathbb{R}^n \rightarrow \{1,2, \dots, K\}$ is a classifier $\rightarrow$?

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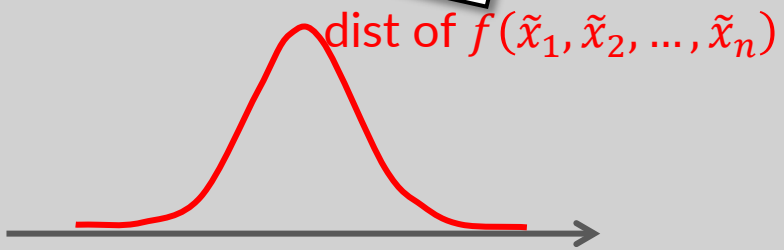

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Data utility by app... error

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Fundamental direction: **Utility analysis** of complex task f

This Proposal: LDP Theory

RIT

- Advancing LDP's mechanism design and utility analysis

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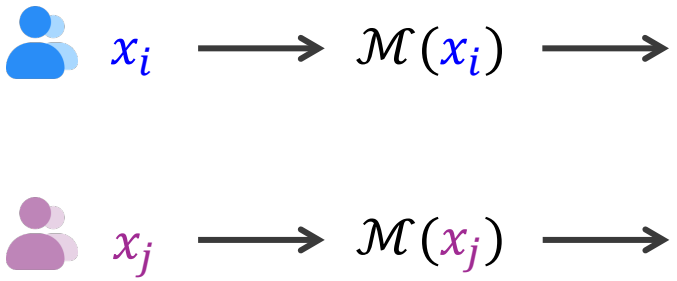
- New LDP mechanisms: (for better privacy
  - correlated LDP mechanisms

Existing LDP mechanisms:
Each user perturbs their data independently



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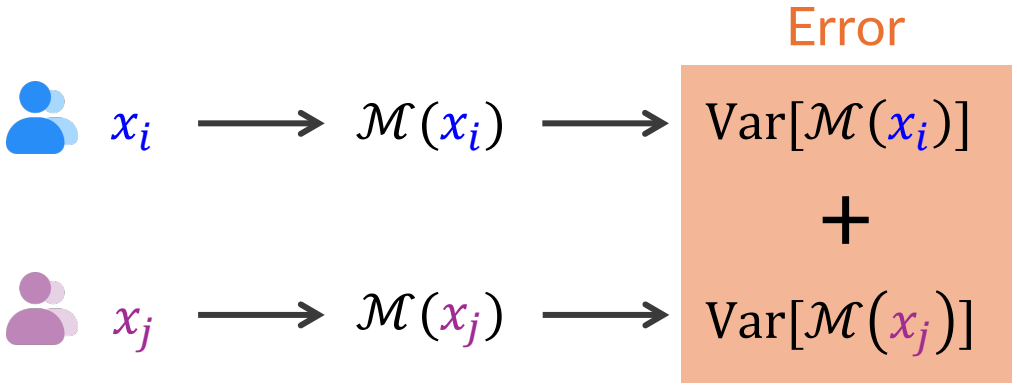

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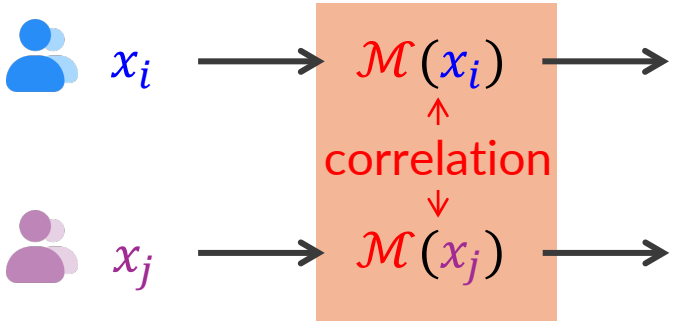
Correlated LDP mechanisms:
Users' data are perturbed in correlated $\mathcal{M}$



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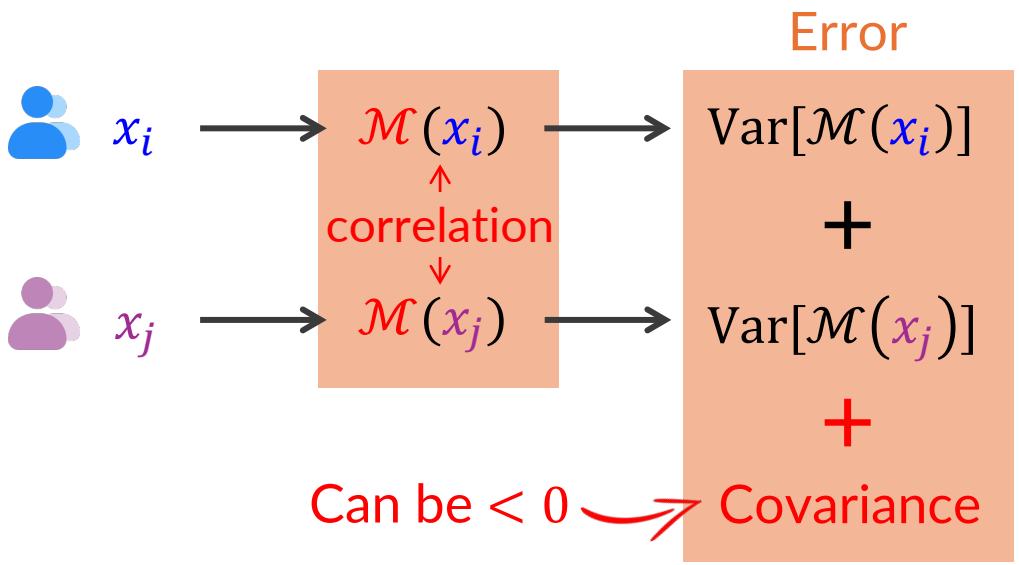

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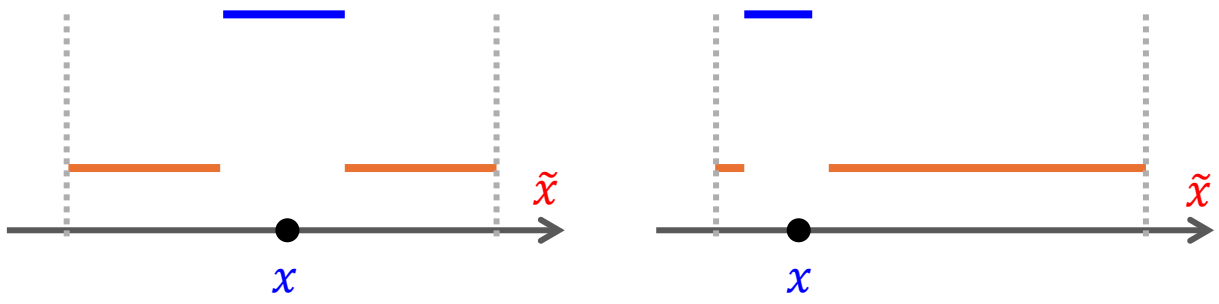
SOTA for bounded numerical data:
Piecewise-based mechanisms (3-piece heuristic PDF)



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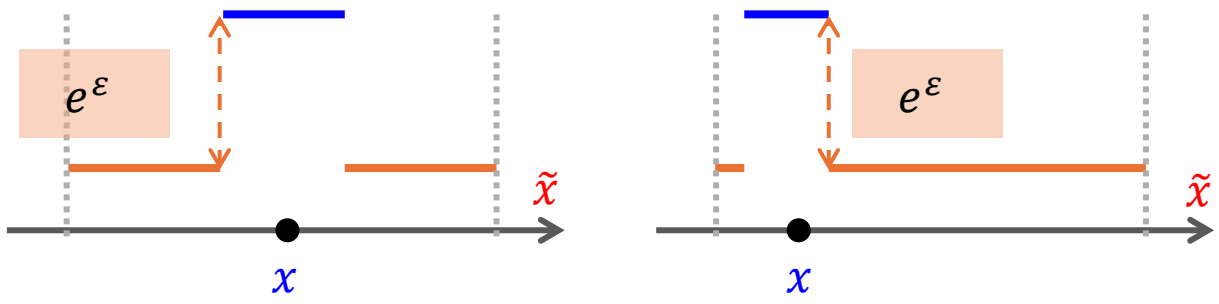

$$\mathrm{pdf}[\mathcal{M}(x) = \tilde{x}] = \begin{cases} p_{\varepsilon} & \text{if } \tilde{x} \in [l_{x,\varepsilon}, r_{x,\varepsilon}] \\ \dfrac{p_{\varepsilon}}{e^{\varepsilon}} & \text{if } \tilde{x} \in \tilde{\mathcal{D}} \backslash [l_{x,\varepsilon}, r_{x,\varepsilon}] \end{cases}$$

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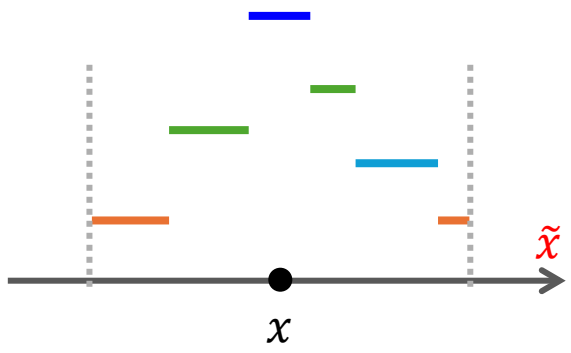
Too heuristic → More generalized version
Potentially has lower error



What is the optimal piecewise-based mechanism?

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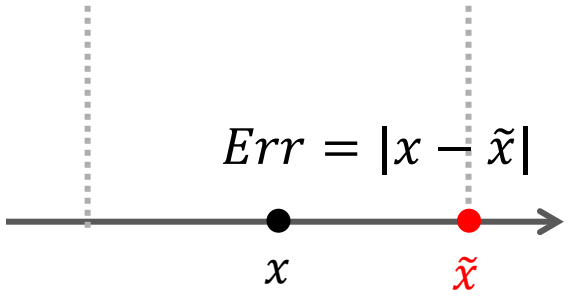
  - optimal piecewise-based mechanisms

Linear data domain → Circular data domain



What is the optimal piecewise-based mechanism?

This Proposal: LDP Theory

RIT

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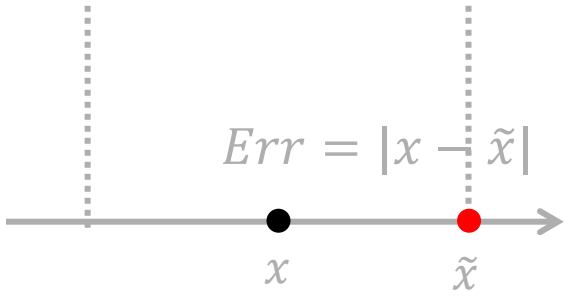

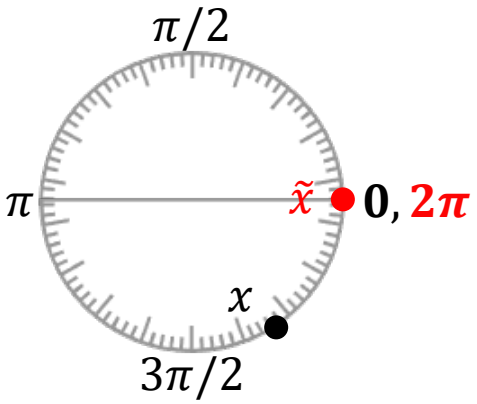

$$Err = \min(|x - \tilde{x}|, |2\pi - x - \tilde{x}|)$$

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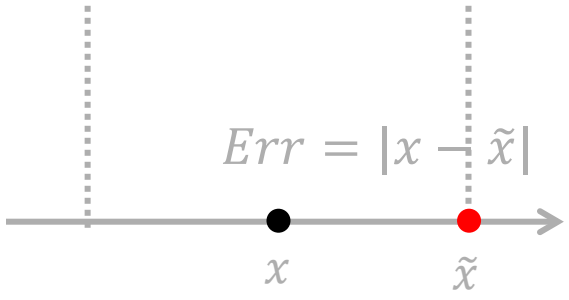

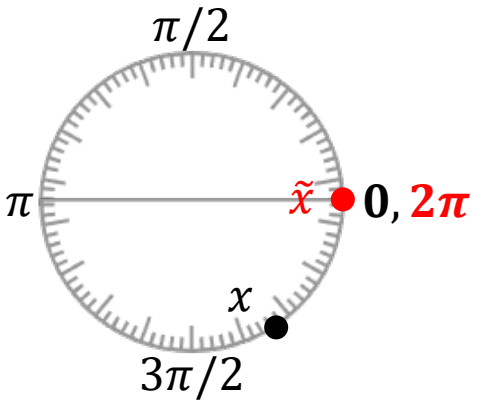

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Optimal piecewise-based mechanism for circular domain?

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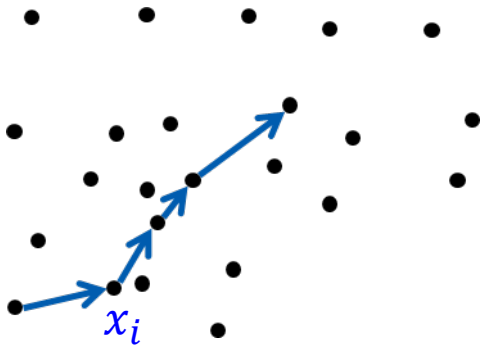
Discrete location space



For $\tilde{x} \in S$

$$\Pr[\mathcal{M}(x_i) = \tilde{x}] \propto e^{d(x_i, \tilde{x})}$$

$$\mathcal{S} = \{p_1, \dots, p_n\}$$

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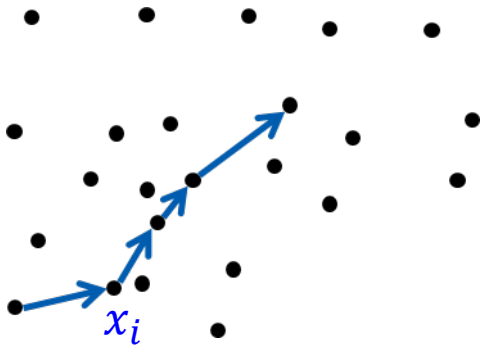

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For $\tilde{x} \in S$

$\Pr[\mathcal{M}(x_i) = \tilde{x}] \propto e^{d(x_i, \tilde{x})}$

Limitations:

- expensive to sample
- only applicable to discrete S

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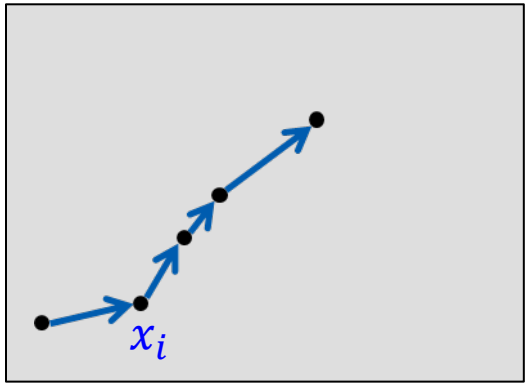
Discrete location space → Continuous location space



$\mathcal{S} = [0,1.5] \times [0,1]$

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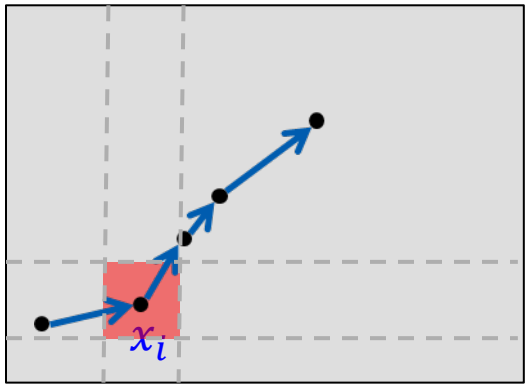

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$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_\varepsilon$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_\varepsilon / e^\varepsilon$

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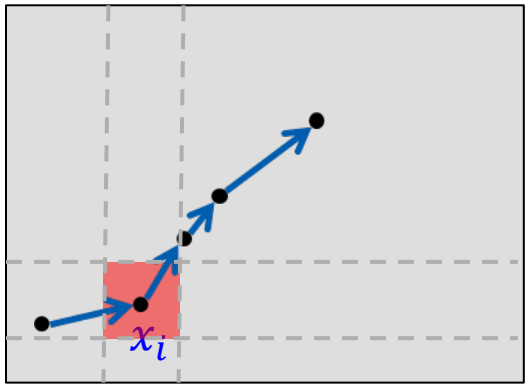

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LDP for continuous S

Benefits:
- negligible sample complexity
- embedded discrete spaces

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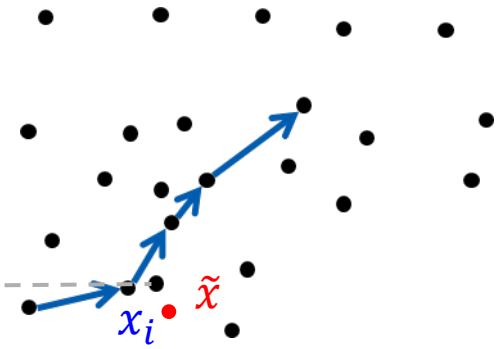
Rounding to nearest
Does not affect privacy

Discrete location space → Continuous location space



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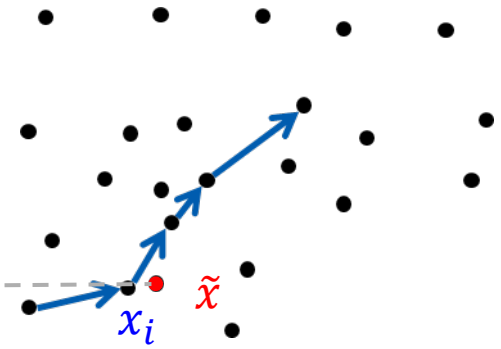

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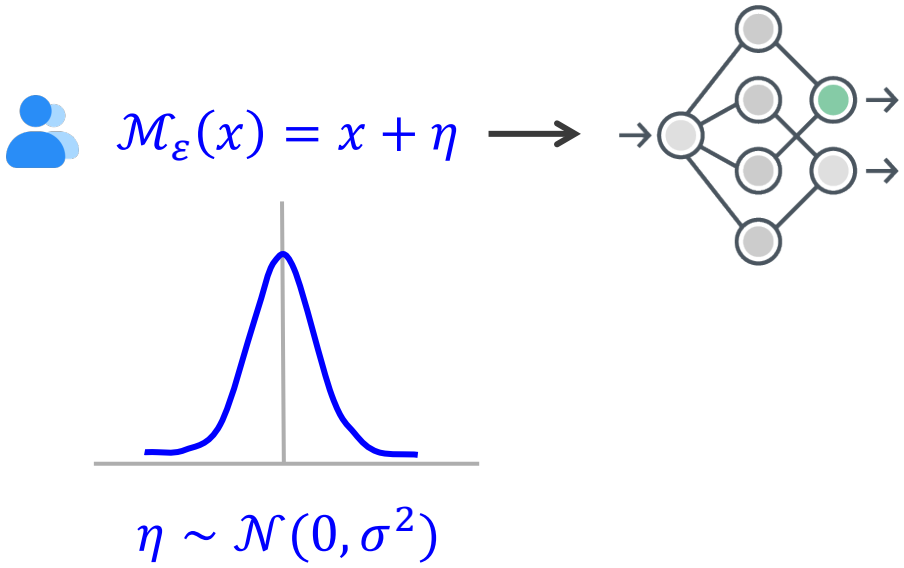
  - classifier utility under LDP-inputs

Empirical classifier utility



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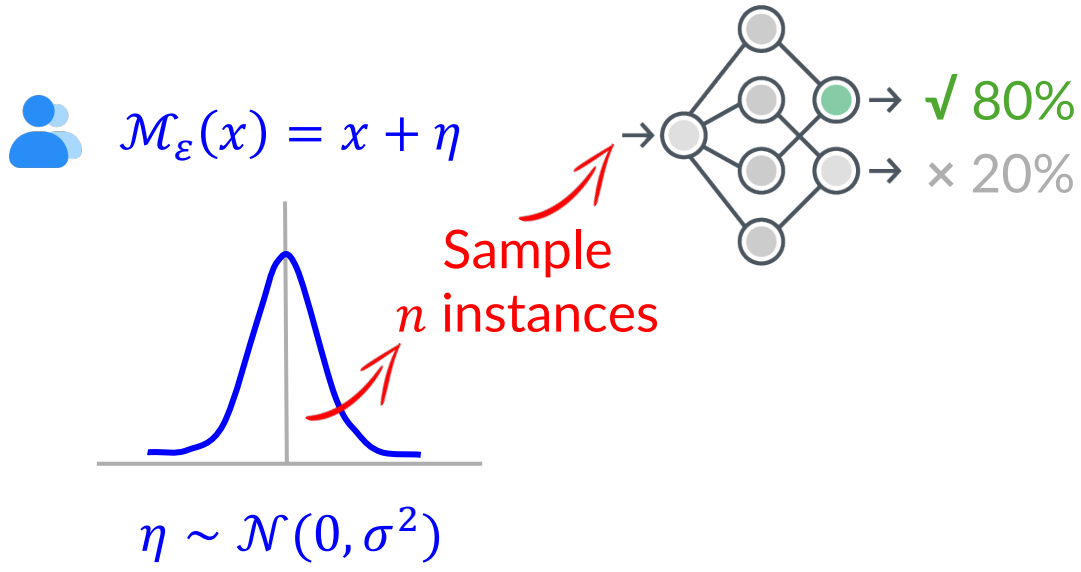

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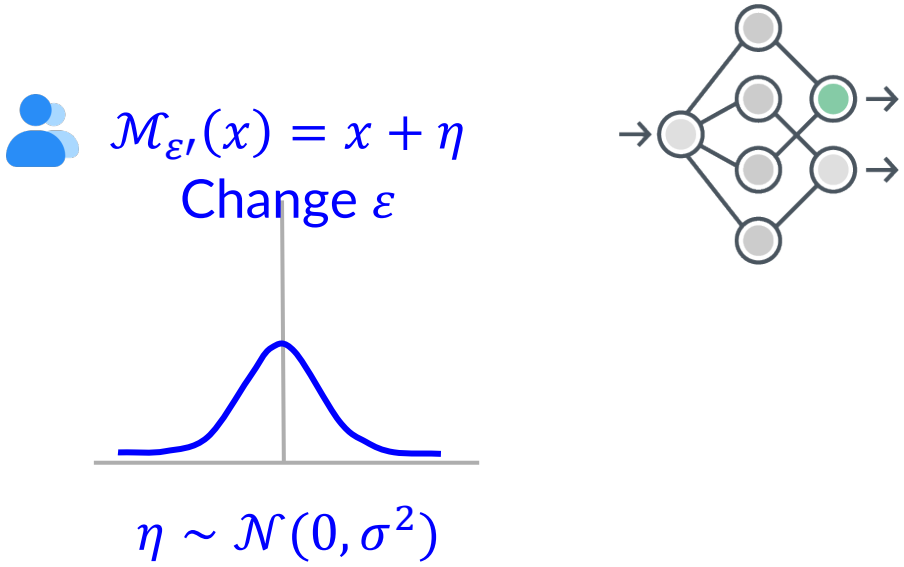

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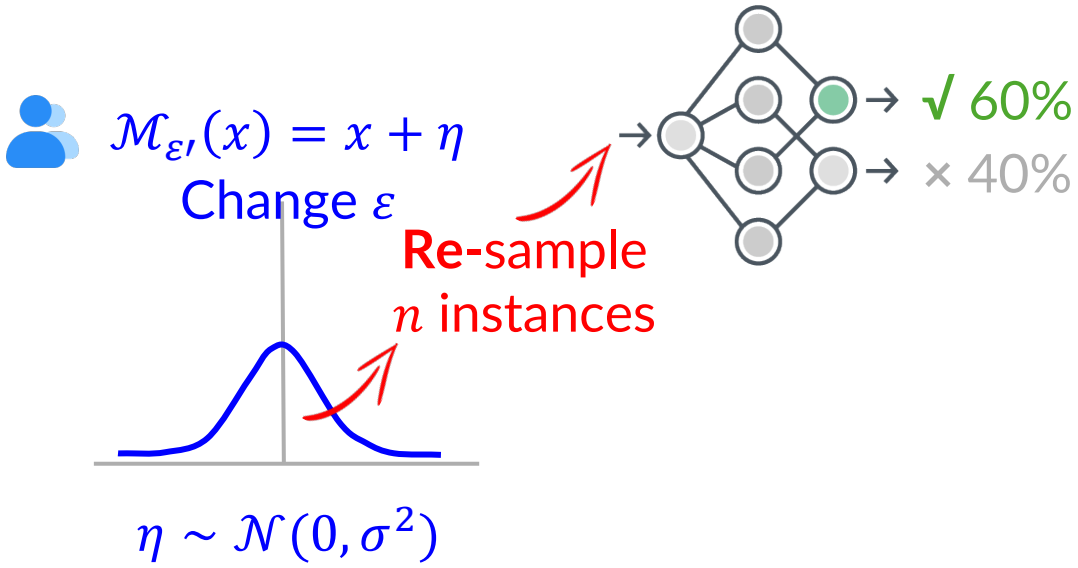

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Empirical classifier utility → Analytical classifier utility

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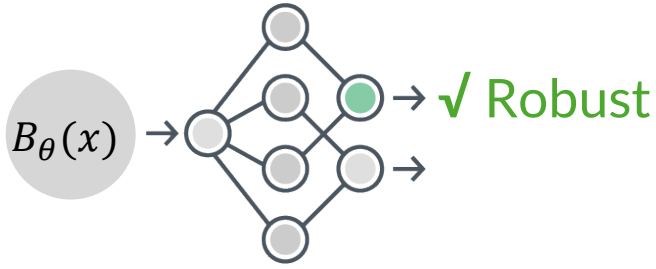

Robustness:
$B_\theta(x)$ is robust

This Proposal: LDP Theory

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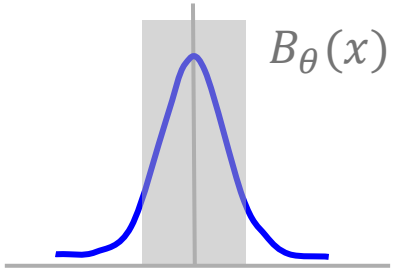
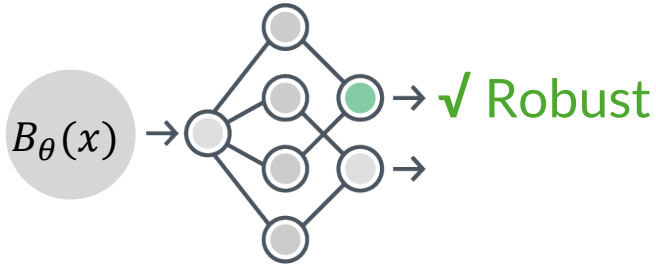
$\mathcal{M}_\varepsilon(x) = x + \eta$





Concentration:
$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)]$
$:= p(\varepsilon, \theta)$

Robustness:
$B_\theta(x)$ is robust

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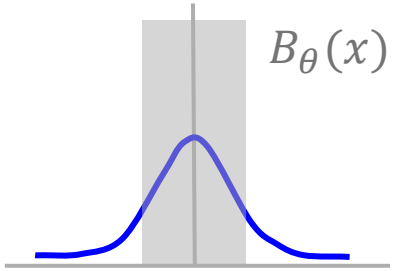

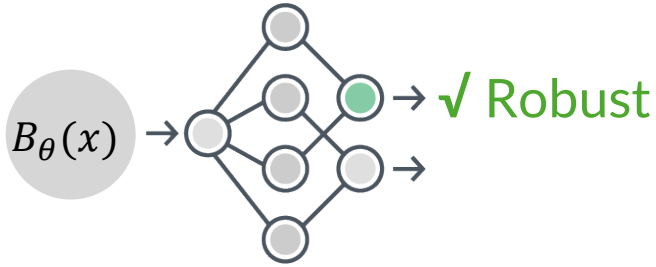

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Connected by θ

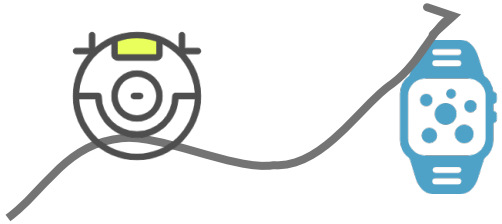
Social Contributions

RIT

- **New LDP building blocks**
  - correlated LDP mechanisms
  - optimal piecewise-based mechanisms

Sensor networks & Federate learning

- **Universal trajectory collection mechanisms**
  - applicable to both continuous / discrete space



Smart home & wearable devices' trajectories

- **Analytical view of classifier utility under LDP-perturbed inputs**
  - choosing ε for when using classifiers